

Bat-Orgil Erdenebat

+1-662-380-7252 | batorgilrb@gmail.com | <https://github.com/BatOrgil7> | <https://linkedin.com/in/bat-orgil-erdenebat>

EDUCATION

University of Mississippi

Oxford, MS

B.S. in Computer Science – Data Science Emphasis; Math Minor; GPA: 3.75

August 2023 - May 2027

Relevant Coursework: Algorithm and Data Structure Analysis, Operating Systems, Architecture of Computer Systems and Assembly, Deep Learning, Machine Learning, Data Science

PROFESSIONAL EXPERIENCE

University of Mississippi – Dept. of Computer & Information Science

Oxford, MS

Teaching Assistant – CSCI 356 & CSCI 111

August 2025 – Present

- Led lab sessions for over **100** Computer Science and STEM majors, reinforcing foundational skills essential to machine learning and AI development.
- Taught **CSCI 356 (Data Structures in Python)**, emphasizing implementation and use of stacks, queues, linked lists, trees, and graphs—key data structures in AI systems and ML pipelines.
- Explained Python's built-in structures (lists, dictionaries, tuples) and their role in real-world data manipulation, feature engineering, and preprocessing tasks.
- Taught **CSCI 111 (Java I)** labs by covering algorithm design, object-oriented programming, and control flow—critical skills for scalable ML/AI system design.
- Provided individualized support to students on debugging and optimization techniques, promoting deeper understanding of algorithmic efficiency and code clarity.

University of Mississippi – Dept. of Computer & Information Science

Oxford, MS

Computer Science Tutor

January 2026 – Present

- Tutored students in programming, data structures, and introductory AI/ML concepts; supported debugging, problem-solving, and exam preparation through one-on-one instruction.

American Cultural & Information Center

Ulaanbaatar, Mongolia

Intern

Sep 2022 - May 2023

- Designed and developed the front-end of the company website using HTML, CSS, and Figma, while also creating social media posters using Canva and Photoshop, contributing to both the website's user experience and the company's digital marketing efforts.
- Expanded outreach efforts by reaching over **356,000** people through targeted social media marketing campaigns, significantly increasing community engagement.
- Developed a comprehensive system design and brand book to ensure the organization's continuous development and consistency in branding and communications.

RESEARCH AND PROJECT

Real-Time ASL-to-Text Recognition System | OpenCV, MediaPipe, TensorFlow, Matplotlib, scikit-learn

- Developed a Computer Vision pipeline using OpenCV and MediaPipe for robust Hand Tracking and Landmark Detection, enabling structured feature extraction from live webcam input.
- Designed and trained a Deep Learning CNN model in TensorFlow/Keras for ASL alphabet classification, achieving 97.19% accuracy on validation data.
- Optimized the system for low-latency Real-Time Inference, integrating preprocessing, model prediction, and text output into a responsive end-to-end deployment pipeline.

CS2 Esports Analytics Platform | SQL, Database Design, Data Modeling

- Engineered normalized relational database and developed complex SQL queries using aggregate functions, nested subqueries, and percentile ranking to identify players; detect outliers in performance distribution
- Designed data model enabling correlation analysis between map-specific performance, team composition factors, and win probability, with schema optimized for regression-ready data extraction and cohort analysis across tournament tiers

CERTIFICATIONS

Career Essentials in Generative AI by Microsoft

Microsoft

Issued June 2024

Ethics in the Age of Generative AI

LinkedIn

Issued June 2024